

Surya Teja Tadaka

Technical Lead | Enterprise AI & Agentic Systems | GenAI • RAG • MCP • Python • AWS

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PROFESSIONAL SUMMARY

Technical Lead with 7+ years of software, data and cloud engineering, specialized in production Generative AI. Designed and led an end-to-end enterprise AI automation platform on AWS Bedrock and Amazon Q Business, using multi-agent orchestration over custom MCP servers to drive Jira tickets through to GitLab merge requests. Built RAG pipelines over thousands of enterprise documents with embeddings and vector search, integrating Claude, Llama and Titan behind reasoning agents. Deep foundation in Python backend engineering, AWS architecture and data platforms — Kafka streaming, Glue ETL and a zero-loss Teradata-to-Redshift migration. Owns engagements end-to-end from technical discovery through production rollout.

SKILLS

AI / Generative AI: Generative AI, LLMs, AI Agents, Agentic Workflows, Multi-Agent Orchestration, RAG, MCP (Model Context Protocol), Tool Calling, Embeddings, Vector Search, Amazon Bedrock, Amazon Q Business, Claude, Llama, Titan, NLP, MLOps, Prompt Engineering

Programming & Backend: Python, TypeScript, JavaScript, SQL, Java, FastAPI, Django, Flask, Node.js, REST APIs, Webhooks, Microservices, Message Queues, Distributed Systems, React

Cloud: AWS (Lambda, ECS, Fargate, API Gateway, S3, DynamoDB, IAM, Redshift, Glue, OpenSearch, SQS, SNS, Step Functions, CloudWatch), GCP, BigQuery, Serverless, Cloud Security

Data: Confluent Kafka, Streaming Pipelines, ETL, Data Pipelines, Data Engineering, Real-Time Processing, Redshift, BigQuery, Parquet, Data Quality Monitoring

Infrastructure & DevOps: Terraform, Infrastructure as Code, Kubernetes, Docker, CI/CD (GitLab, GitHub Actions, Jenkins), Observability (Grafana, CloudWatch, DataDog), Production Debugging

Customer & Delivery: Technical Discovery, Scoping, Solution Architecture, Stakeholder Alignment, Demo-Driven Delivery, Forward Deployment, Project Leadership, Mentoring

PROFESSIONAL EXPERIENCE

Southwest Airlines (Qentelli) — Tech Lead

TX | June 2023 – Present

- Designed and led development of an end-to-end AI automation platform using AWS Bedrock, Amazon Q Business and multi-agent orchestration to auto-generate code, create branches and raise GitLab merge requests directly from Jira tickets.
- Built a multi-agent ecosystem spanning Jira, GitLab and Xray through custom MCP (Model Context Protocol) servers, enabling structured tool-calling and cross-system automation.
- Architected a scalable RAG pipeline using embeddings and vector databases to ingest and retrieve thousands of Confluence documents, enabling Bedrock models to answer engineering questions accurately.
- Integrated Bedrock LLMs (Claude, Llama, Titan) with reasoning agents to improve code generation, ticket comprehension and automated documentation generation.
- Ensured enterprise reliability through rate limiting, error handling, observability, fallback logic and structured agent workflows.
- Delivered the platform end-to-end: technical discovery, requirements gathering, architecture, development, CI/CD, AWS deployment, monitoring and iterative optimization.
- Architected production backend services and automation workflows using Python, FastAPI, React, Docker, Terraform and AWS; mentored junior engineers and established practices around design, observability and reliability.

Smile Direct Club (Qentelli) — Sr. Software Engineer

Remote, USA | Oct 2020 – May 2023

- Led successful migration from on-prem Teradata to AWS Redshift ensuring zero data loss and optimized performance.
- Architected and deployed Confluent Kafka on AWS for real-time streaming and event-driven data processing.
- Automated AWS Glue ETL workflows to transform and store data in S3 (Parquet), improving reliability and reducing manual operations.
- Integrated AWS services including Lambda, DynamoDB, OpenSearch, Step Functions, SQS, SNS and S3 using boto3; designed batch and real-time pipelines in Python and Java.
- Built data quality monitoring using SonarQube and DataDog, reducing pipeline failures and downtime by 30%; achieved 95% automated test coverage for ETL processes.
- Converted manual infrastructure deployments to Terraform and YAML IaC, reducing environment setup time by 50%.
- Reduced cloud infrastructure cost by \$100K annually through workflow automation and test/reporting optimizations.

SS&C Technologies — Software Engineer

India | May 2019 – April 2020

- Developed backend services using Django, Flask, React and Spring Boot for claims adjudication and health insurance workflows.
- Implemented secure authentication and data protection using OAuth2, SSL, JWT and encryption.
- Applied BeautifulSoup and NLTK for parsing and text extraction, improving data accuracy and automated claims adjudication.
- Built containerized applications using Docker and CI/CD workflows using Jenkins to automate builds, tests and deployments.
- Built automated data backup pipelines to AWS S3, ensuring compliant storage of claims data.
- Reduced production defects by enforcing automated test coverage and environment parity.

SELECTED PROJECTS

Enterprise AI Code-Automation Platform

Multi-agent platform on AWS Bedrock and Amazon Q Business that takes a Jira ticket through branch creation, code generation and an open GitLab merge request. Agents reach Jira, GitLab and Xray through custom MCP servers; hardened with rate limiting, fallback logic and observability. (Southwest Airlines)

Confluence RAG Knowledge Assistant

Retrieval-augmented generation pipeline over thousands of Confluence documents — ingestion, chunking, embeddings and vector retrieval feeding Bedrock models to answer internal engineering questions with grounded citations. (Southwest Airlines)

Teradata → Redshift Migration & Streaming Platform

Zero-data-loss warehouse migration paired with a Confluent Kafka streaming backbone and automated Glue ETL into S3/Parquet; 95% ETL test coverage, 30% fewer pipeline failures, \$100K annual cloud cost reduction. (Smile Direct Club)

Cloud-Native Infrastructure as Code — github.com/TadakaSuryaTeja/aws_eks

Declarative Amazon EKS provisioning with Terraform and eksctl for repeatable, version-controlled Kubernetes environments.

Broadcom Driver Installer — github.com/TadakaSuryaTeja/BroadcomInstaller2021

Open-source Linux hardware-enablement tool adopted by the community (42 stars, 27 forks).

EDUCATION

Vardhaman College of Engineering — Bachelor of Technology in Computer Science | Graduation: 2019

Government Polytechnic — Diploma in Computer Science | Graduation: 2016

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate — Amazon Web Services
- AWS Certified Cloud Practitioner — Amazon Web Services
- Machine Learning and Artificial Intelligence — AppliedAI
- Information Security-IV — NPTEL
- Mobile Application Development — NPTEL